Brier score (i.e., the standard ℓ_2 loss). Our goal is not to produce the best possible ensemble forecast, but rather to derive interpretable coefficients that serve as proxies for the relative information each forecast provides about the event of interest. We report point estimates for these coefficients along with confidence intervals computed from 1000 bootstrap samples. To contextualize these results, we include the Brier score achieved by a leave-one-out version of this simplex regression procedure, providing an out-of-sample estimate of the ensemble's performance.

	AIA Forecaster		**Market**		**Ensemble**
Benchmark	Brier score	Coefficient	Brier score	Coefficient	Brier score
FB-Market	**0.075**	0.87 [0.42, 1.00]	0.097	0.14 [0.00, 0.58]	0.079
MarketLiquid	0.126	0.33 [0.12, 0.47]	0.111	0.67 [0.53, 0.80]	**0.106**

Table 7: The performance of a simplex-constrained regression ensemble which combines LLM forecasts with the market price. Regression coefficients are reported with 95% bootstrap confidence intervals over 1000 samples. Ensemble performance is reported as a leave-one-out estimate. Coefficients may not sum to 1 due to rounding.

As Table 7 demonstrates, the AIA Forecaster is assigned the majority of the ensemble weight on the FB-Market benchmark, and the leave-one-out ensemble performs worse than the AIA Forecaster alone. Thus, the market price does *not* provide signal that is additive to the forecasts produced by our system. However, on MarketLiquid, where the market price outperforms the AIA Forecaster, the converse is not true. Instead, the ensemble assigns roughly a third of its weight to the LLM forecasts, and the leave-one-out ensemble outperforms both the LLM forecasts and the market price. This suggests that even in challenging settings where LLMs fail to outperform the market consensus outright, they can still provide valuable and diversifying information to decision makers.

6 Search Is Critical for Live Markets

We now turn to evaluating the AIA Forecaster on prediction markets which have not yet resolved ("live markets") at the time of forecaster execution. This kind of evaluation precludes any form of foreknowledge bias, as we are asking the model to predict events which have not yet occurred.

To do this, we collected a dataset of live markets on a popular prediction market platform, filtered by relevance and liquidity. Our filtering process for the live markets was the same process used to generate the static MarketLiquid dataset (except filtered to markets that are currently open). We conducted this process nightly between August 15, 2025 and August 21, 2025, which yielded approximately 1500 questions per night. We sampled to 250 questions at random every night to generate the evaluation set. These questions changed on a nightly basis as new markets opened and existing markets were resolved.

We conduct two sets of experiments. For both experiments, we executed the AIA Forecaster with and without search enabled. In both conditions, we omitted the market prices to isolate the effect of search on forecaster performance.[13]

Market Type	**With Search** (Brier score)	**Without Search** (Brier score)	**Market Baseline** (Brier score)
Closed markets (n=64)	**0.1002**	0.3609	0.1111
Open markets (n=1750)	**0.0522**	0.0910	—

Table 8: Real-time forecasting performance with and without search. In both cases the system is prevented from accessing market prices. In closed markets, forecasts are evaluated against ground truth outcomes. In open markets, forecasts are scored against market prices. The performance of the market consensus on closed markets is included for reference. AIA Forecaster improves on the no-search baseline by 3.6× in closed markets and is more concordant with market prices in open markets.

In the first experiment, we evaluate the Brier score on the 64 markets that closed by August 26, 2025. As shown in Table 8, with search, the AIA Forecaster achieved a Brier score of 0.1002. Without search, the AIA Forecaster achieved a Brier score of 0.3609, a difference of 3.6×. As we can see, search is critical for forecasting; in fact, excluding search yields performance which is worse than uniformly predicting 50% (which mechanically achieves a Brier score of 0.25).

We also compute the Brier score achieved by simply using the market price as a baseline ("consensus") forecast. We see that, with search, the AIA Forecaster slightly outperforms the market consensus, although with only 64 markets we are unable to statistically distinguish these two results.

In the second experiment, we evaluate the Brier score *with respect to the market price*. Unlike the Brier score for resolved markets, a perfectly skilled forecaster would score substantially away from 0. Nonetheless, market prices are a strong baseline forecast (Surowiecki, 2004; Bassamboo et al., 2015; Da and Huang, 2019; Lichtendahl Jr et al., 2013), so significant deviations from a market price baseline are unlikely for a skilled forecaster. Importantly, this allows us to evaluate the AIA Forecaster over the much larger (1750) set of markets which are still open.

As shown in Table 8, compared to market prices, AIA Forecaster achieves a Brier score of 0.0522 with search enabled. Without search, our forecaster achieves a Brier score of 0.0910. Because we expect market prices to be informative, being close to the market price likely indicates better performance. Thus, we can again see that search is critical for generating informative forecasts.

These experiments on live markets are necessarily narrower in scope than the larger-scale retrospective evaluations we conduct in Section 4. Nonetheless, we again see evidence

13. As in Section 5.1, we exclude market prices from the prompt and prevent the forecaster from accessing domains associated with the most popular prediction market platforms.

which corroborates our main findings, and further suggests that foreknowledge bias is not meaningfully impacting our results.

7 Ensembling, Synthesis, and Statistical Corrections

In this section, we explore various approaches to ensembling multiple forecasts into a stronger aggregate forecast. We find that some form of ensembling is critical and dramatically outperforms naive baselines. In Section 7.1, we motivate the need for ensembling forecasts, and find that simple averaging of independent forecasts (and minor variations thereof) dramatically outperforms a naive baseline which eschews any form of aggregation. In Section 7.2, we find that an additional "supervisor" agent can substantially improve on the performance of simple mean aggregation by actively resolving disagreements between individual forecasts. Finally, in Section 7.3, we find that the additional application of Platt scaling improves our performance to match that of superforecasters.

7.1 Simple Averaging Is a Strong Baseline

We now explore the performance of simple aggregation methods. First, to motivate the need for aggregating multiple forecasts, consider the following simple model. Let the random variables $Q \in \mathcal{Q}, P \in [0,1], O \in \{0,1\}$ denote questions, forecasts and outcomes, respectively. Let (Q, P, O) be a random vector with some well-defined probability measure. Let $\ell(P, O) = (O - P)^2$ be the Brier score. Under this model, fixing a question $Q = q$ induces some conditional distribution over the question-specific Brier score ℓ. Because ℓ is strongly convex, the following is an immediate consequence of Jensen's inequality:

$$\ell\left(\mathbb{E}[P \mid Q = q], \mathbb{E}[O \mid Q = q]\right) < \mathbb{E}\left[\ell\left(P, O\right) \mid Q = q\right] \qquad \forall q. \tag{1}$$

That is, the Brier score for the average forecast and outcome is a strict improvement on the Brier score averaged over possible forecasts and outcomes.[14] For example, given the choice between (1) predicting exactly 50% or (2) predicting 100% and 0% with equal probability, the former is preferable *regardless of the true outcome*. This suggests that averaging many independent forecasts for a given question will tend to improve on sampling a single forecast. We confirm this empirically in Figure 3 below, which demonstrates sharply decreasing Brier scores on MarketLiquid as we increase the number of independent forecasts from 1 to 5, and an additional modest improvement as we further increase the number to 15 forecasts.[15]

14. Breiman (1996) uses this fact to motivate ensembling in the classical statistical learning paradigm.
15. To maintain computational tractability, Figure 3 presents the results of bootstrap resampling a fixed number of forecasts from superpopulation of 50 forecasts per question rather than repeatedly re-evaluating the benchmark at each point on the x-axis. We use the larger MarketLiquid benchmark to increase statistical power.